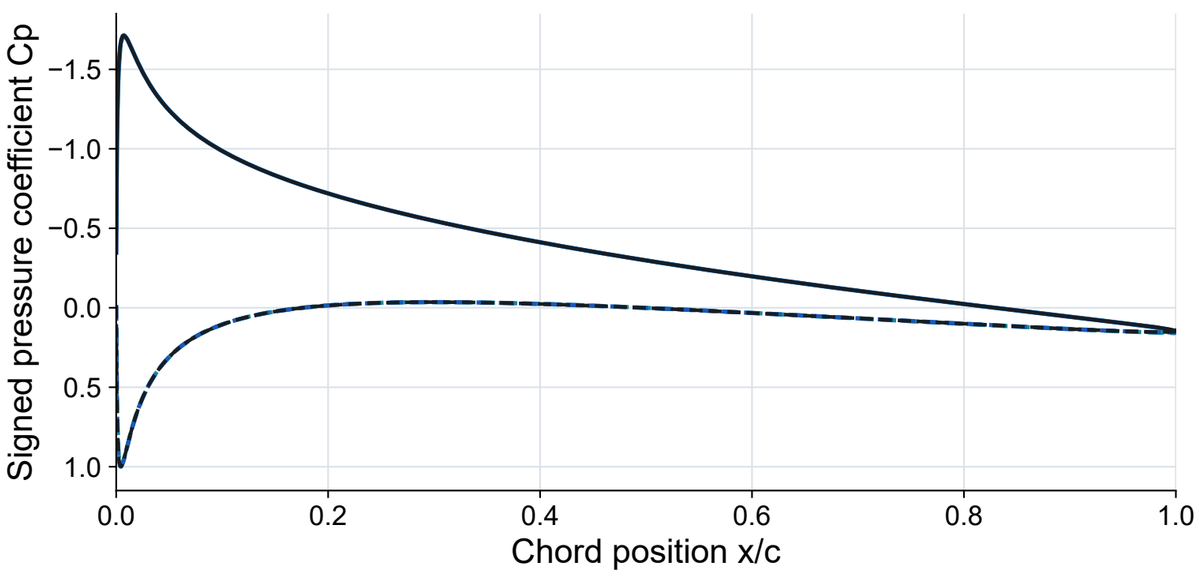
Independent 2D airfoil verification · α = 4°

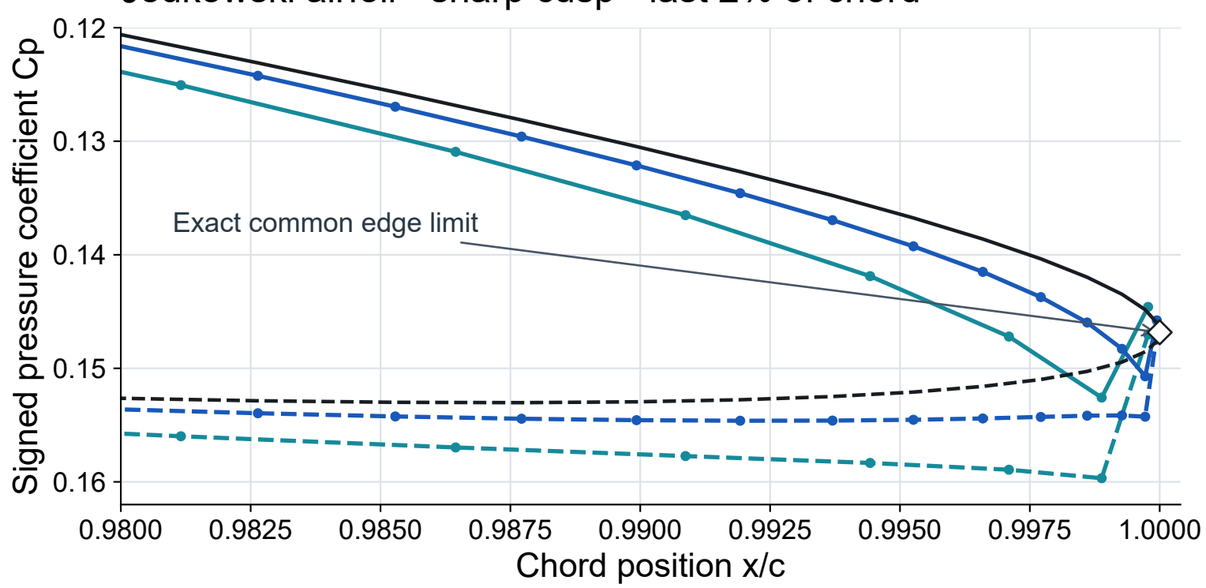
Every curve uses the same stated conformal airfoil and angle. Cp is compared at matching physical points.

Current solver · 160 contour panels
Current solver · 320 contour panels
Exact potential flow
Upper surface: solid
Lower surface: dashed

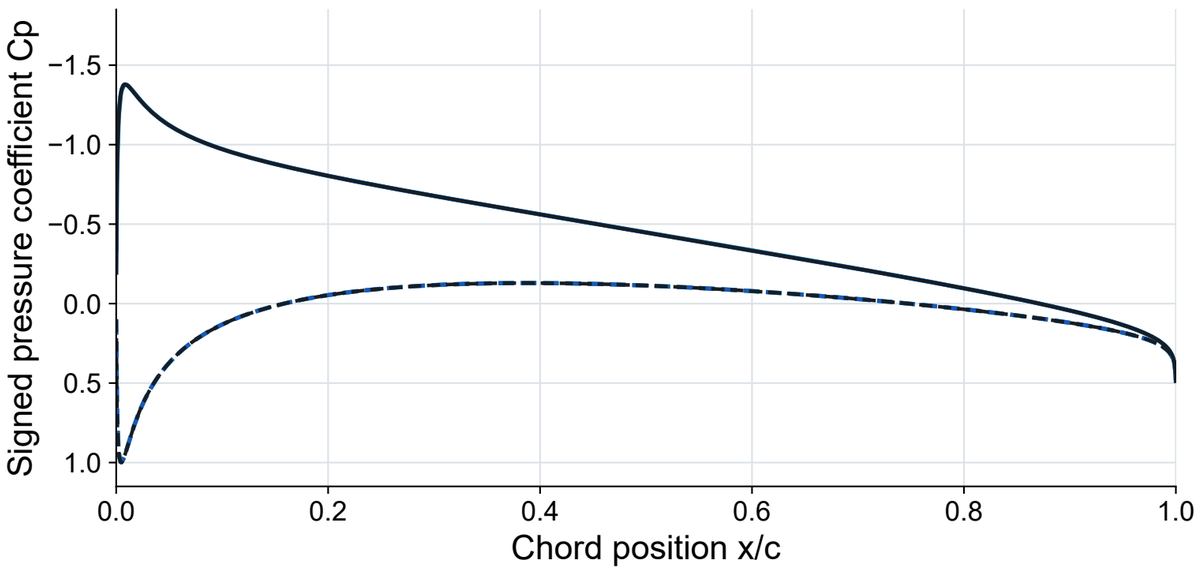
Joukowski airfoil · sharp cusp · full chord



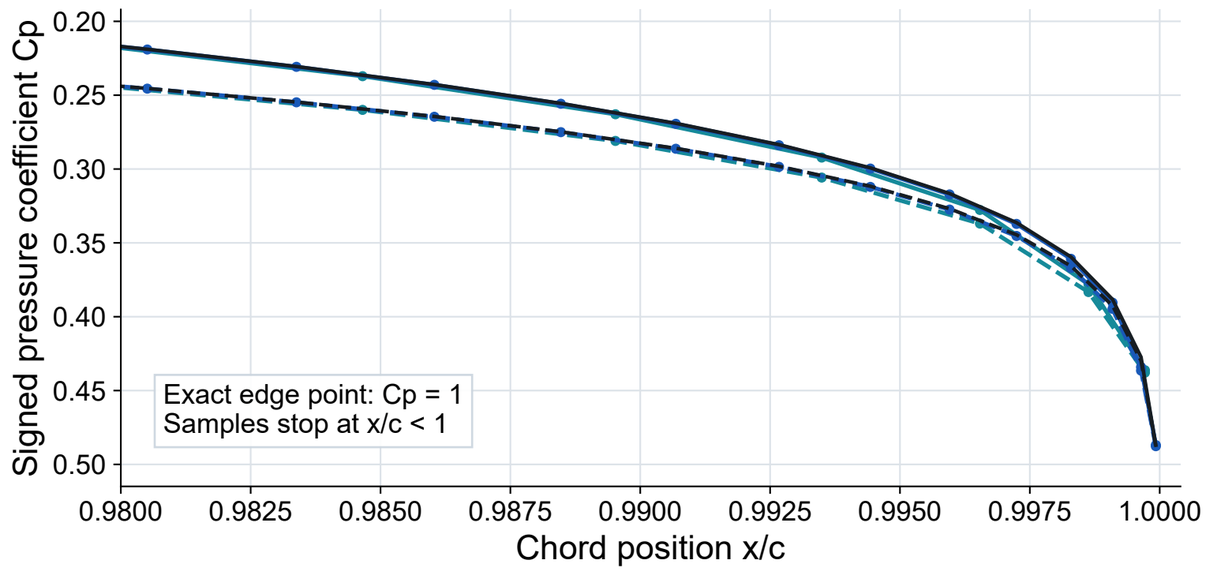
Joukowski airfoil · sharp cusp · last 2% of chord



Karman–Trefftz airfoil · 12° finite edge · full chord



Karman–Trefftz airfoil · 12° finite edge · last 2% of chord



Dots are panel-centre predictions, not edge values. The near-edge bends are retained numerical errors, not physical shocks.

Sharp cusp: max |ΔCp| among the last 3 panels = 0.0100 (160) and 0.00587 (320). See the separate signed-error figure.

Source: committed TE_PROFILES.csv and TE_REPAIR_REVIEW.json. No clipping, smoothing, reference replacement or new solver run.